



FlowForge v1.0

User Manual

Centrifugal Pump Preliminary Design Tool

Version	Method	Platform
v1.0	1D Meanline	Web (HTML)

1D Meanline Preliminary Design · Multi-Case · Performance Curves
Version 1.0 · Browser-based, no installation required

1. Overview

FlowForge v1.0 is a browser-based 1D meanline tool for the preliminary hydraulic design of centrifugal pump impellers. It requires no installation — simply open the HTML file in any modern web browser (Chrome, Firefox, Edge, Safari).

The solver implements the classical Euler turbomachinery equation combined with four slip-factor models and geometry-sensitive loss correlations. Starting from three operating-point inputs (flow rate Q , target head H , and rotational speed n), it iterates on impeller outer diameter D_2 until the delivered net head converges to the target within the specified tolerance.

Version 1.0 adds a full off-design Performance Curve module, allowing the user to plot $H(Q)$, $\eta(Q)$, $P_{\text{shaft}}(Q)$, and $\text{Incidence}(Q)$ curves over any flow-rate range after the design point is solved.

ⓘ The tool is for preliminary design only. It uses 1D meanline assumptions and does not replace 3D CFD or detailed mechanical design. All results should be validated against pump test data or higher-fidelity models before manufacturing.

1.1 Key Capabilities

- Up to 8 simultaneous design cases with tab-based navigation and side-by-side comparison table
- Four slip-factor models: Stodola, Wiesner (1967), Pfleiderer (1961), and user-specified constant σ
- Automatic Wiesner validity check with fallback to Stodola when $\phi \geq \phi_{\text{lim}}$
- Dual specific-speed reporting: ns_{design} (from inputs) and $ns_{\text{delivered}}$ (from converged geometry)
- Geometry-sensitive hydraulic loss model: friction (k_h), incidence (k_i), and disk-friction (k_{df})
- Inlet velocity triangle: auto or manual D_1 and b_1 , pre-swirl Vu_1 support
- km Sensitivity analysis: dD_2/dkm at constant H , n , Q
- km vs ns consistency check against Gülich (2020) regime table
- Performance Curve module: $H(Q)$, $\eta(Q)$, $P(Q)$, $\text{Incidence}(Q)$ with BEP marker
- CSV export of all converged cases
- Bilingual interface: English / Turkish toggle

2. Getting Started

⚠ Firefox File Protocol Restriction: When opened via the file:// protocol (double-click on most systems), Firefox may block certain JavaScript features required by the solver due to cross-origin security policies. If the interface does not respond or inputs are not processed, serve the file through a local web server instead: run `python3 -m http.server 8080` in the file's directory, then navigate to `http://localhost:8080/FlowForge_v1.0.html` in Firefox. Chrome and Edge do not have this restriction.

Open FlowForge v1.0.html in a modern web browser by double-clicking it or dragging it into the browser window. No internet connection is required — all calculations run locally in JavaScript.

The interface language defaults to English. To switch to Turkish, click the TR button in the top-right corner of the header bar. All labels, notes, and status messages will update instantly.

2.1 Workflow Summary

- Step 1 — Set the number of cases (1–8) and click Apply.
- Step 2 — Enter operating-point inputs for the active case: Q , H_{target} , n .
- Step 3 — Select a slip model (Stodola is recommended as the default).
- Step 4 — Set design parameters: β_2 , km , η .
- Step 5 — Optionally adjust Fluid & Inlet and Solver Settings.
- Step 6 — Click Run Case. Review results, warnings, and the iteration log.
- Step 7 — (Optional) Compute performance curves using the Performance Curves panel.
- Step 8 — Add more cases, run all, compare, and export CSV if needed.

① For a quick first run, leave all fields at their default values and click Run Case. The defaults represent a typical industrial pump: $Q = 0.05 \text{ m}^3/\text{s}$, $H = 40 \text{ m}$, $n = 2900 \text{ rpm}$, Stodola slip model.

2.2 Worked Example — Default Inputs

Running the solver with all default values ($Q = 0.05 \text{ m}^3/\text{s}$, $H = 40 \text{ m}$, $n = 2900 \text{ rpm}$, Stodola slip model, $\beta_2 = 25^\circ$, $km = 0.15$, $\eta = 0.75$, $Z = 7$) produces the following converged results after 7 iterations:

Output	Symbol	Value	Unit
Outer diameter	D_2	177.4	mm
Outlet width	b_2	24.1	mm
Width ratio	b_2/D_2	0.136	—
Blade tip speed	u_2	26.94	m/s
Slip factor (Stodola)	σ	0.8103	—
Net head (converged)	H_{net}	39.69	m
Shaft power (η override)	P_{shaft}	25.91	kW
Specific speed (design)	ns	40.3	— (SI, dimensionless)
Specific speed (delivered)	ns_{del}	41.0	—
Shape number	Ω_s	0.775	—

① The solver converges to $H_{\text{net}} \approx 39.69 \text{ m}$ ($\approx 0.8\%$ below the 40 m target), which is within the default 1 % tolerance. To tighten convergence, reduce the tolerance from 0.01 to 0.005 in Solver Settings.

3. Input Parameters

3.1 Operating Point (Required)

These three inputs define the hydraulic duty. The solver will size D_2 to meet this duty.

Parameter	Default	Range	Notes
Q	0.05 m ³ /s	> 0	Flow rate. Select unit from dropdown: m ³ /s, L/s, L/min, L/h, m ³ /h, US gal/min, kg/s.
H_target	40 m	> 0	Target net head in metres. The solver iterates until $H_{net} \approx H_{target}$.
n	2900 rpm	> 0	Rotational speed. Directly sets blade tip speed $u_2 = \pi \cdot D_2 \cdot n / 60$.

① When using kg/s as the flow unit, ensure the fluid density ρ is set correctly in the Fluid & Inlet card — the conversion to m³/s depends on ρ .

3.2 Slip Model

Slip factor σ accounts for the deviation of the actual exit whirl velocity from the theoretical (ideal) value. The four available models are:

- **Stodola (default):** $\sigma = 1 - (\pi/Z) \cdot \sin(\beta_2)$. Valid for the full range of ϕ . Recommended starting point.
- **Wiesner (1967):** $\sigma = 1 - \sqrt{(\sin \beta_2) / Z^{0.7}}$. More accurate for typical pump geometry. Automatically falls back to Stodola when $\phi = \phi_{lim}$ exceeds the validity limit ϕ_{lim} .
- **Pfleiderer (1961):** $\sigma = 1 / (1 + p \cdot \phi / \tan \beta_2)$, where $p = 0.65 \cdot (1 + \sin \beta_2)$. Captures the influence of meridional velocity ratio more explicitly.
- **Constant σ :** User-specified value (0.65–0.97). Use only when σ is known from test data.

① Stodola and Wiesner assume zero pre-swirl ($Vu_1 = 0$, $\alpha_1 = 90^\circ$). If pre-swirl is non-zero, Pfleiderer or Constant σ is more appropriate.

3.3 Design Parameters (Fixed — Not Iterated)

These values are held constant during the D_2 iteration.

Parameter	Default	Range	Notes
β_2	25°	[15, 45]°	Outlet blade angle from meridional direction. Lower values give higher head and higher losses. Typical industrial range: 20–35°.
km	0.15	[0.05, 0.50]	Meridional velocity ratio: $km = Vm_2/u_2$. Governs the meridional velocity at the impeller exit. Typical: 0.10–0.25.
η	0.75	(0, 1.0]	Overall efficiency override. Set $\eta = 0$ to activate the physics-based loss model. If non-zero, the physics-based loss model is bypassed and η is imposed directly.
Z	7	[2, 30]	Number of impeller blades. Odd numbers preferred to avoid resonance. Typical: 5–9.

⚠️ η Override — Important: When $\eta \neq 0$, the physics-based loss model (k_h , k_i , k_{df}) is bypassed entirely. The solver imposes the supplied η directly and does not predict losses from geometry. This disables the tool's predictive capability. Use η override only when efficiency is known from test data or a validated reference pump of similar specific speed and geometry.

3.3.1 Loss Coefficients k_h , k_i , k_{df}

Three lumped loss coefficients control the hydraulic loss model when $\eta = 0$ is selected (physics-based mode). They appear in the Advanced / Loss Coefficients card of the interface.

Coefficient	Default	Typical Range	Description
k_h	0.15	0.05 – 0.25	Hydraulic friction coefficient. Scales the Euler head: $H_f = k_h \cdot H_{\text{euler}}$. Represents passage friction, leakage, and recirculation losses in the impeller channels.
k_i	0.50	0.20 – 1.00	Incidence loss coefficient. Scales the incidence penalty: $H_i = k_i \cdot (V_{m1}^2/2g) \cdot i^2$. Increase for blunt leading edges or large incidence excursions.
k_{df}	0.0015	0.0005 – 0.005	Disk-friction coefficient. Scales the disk-friction power: $H_{df} = k_{df} \cdot u_2^2/(2g) \cdot D_2$. More sensitive to tip speed than to flow rate.

① The defaults ($k_h = 0.15$, $k_i = 0.5$, $k_{df} = 0.0015$) are calibrated for well-designed industrial impellers at high Reynolds numbers and water-like fluid viscosities. For worn impellers or rough castings, raising k_h to 0.20–0.25 is a reasonable first adjustment.

⚠️ Note on k_{df} (Disk Friction): The disk-friction coefficient k_{df} is calibrated for water-like fluids ($\mu \approx 0.001 \text{ Pa}\cdot\text{s}$) at high Reynolds numbers and implicitly assumes turbulent boundary layer behaviour on the shroud and hub faces. It does not scale with viscosity in the current implementation. For fluids of elevated dynamic viscosity ($\mu > \sim 5 \text{ mPa}\cdot\text{s}$), the model will underpredict disk friction losses — k_{df} should be increased accordingly, or the result interpreted as a lower-bound estimate.

3.4 Fluid & Inlet (Supplementary)

FlowForge v1.0 introduces an integrated fluid property tool in the Fluid & Inlet card. Instead of entering a density value manually, the user selects the working fluid and operating temperature; ρ is computed automatically.

3.4.1 Fluid Type

Two options are available via radio buttons:

- Water** — density calculated from a polynomial fit to NIST data [7] (valid 0–100 °C, max error < 0.05 kg/m³).
- EG / Water** — density read directly from the engine coolant property database [8] for ethylene-glycol / water mixtures. No interpolation is performed; only the exact table values are used.

3.4.2 Temperature Selection

Temperature is selected by toggle (radio buttons), not typed freely. Available values for both fluid types:

Fluid	Available temperatures (°C)
Water	0 · 20 · 40 · 60 · 80 · 100
EG / Water	0 · 20 · 40 · 60 · 80 · 100

① The EG/Water temperature toggle is restricted to these six values because the source dataset [8] provides measured density at these exact temperatures only. Free-entry is intentionally disabled for EG/Water to prevent incorrect values from entering the solver.

3.4.3 EG Concentration (EG / Water only)

When EG / Water is selected, a second toggle appears for EG concentration by volume:

Concentration	Density range (kg/m³)	Typical application
30 %	990 – 1060	Mild climate / light frost protection
50 %	1025 – 1090	Standard automotive / heavy-duty engine coolant
70 %	1055 – 1130	Severe cold climate / high antifreeze requirement

① Only 30 %, 50 %, and 70 % are available — these are the three concentrations present in the source dataset [8]. Intermediate values (e.g. 40 %, 60 %) are not supported; select the nearest value or enter ρ manually.

3.4.4 Density Field Behaviour

The ρ field updates automatically whenever the fluid type, temperature, or concentration toggle changes. The label shows (auto) when the value is computed. If the user edits ρ directly, the label switches to (manual) and automatic updates pause until the fluid type or concentration toggle changes again.

① For fluids not covered by the table (e.g. oil, other glycols, seawater), select any fluid type for an approximate starting value, then manually overwrite ρ . The (manual) label confirms the override is active.

3.4.5 Full EG / Water Density Table

Reference values used by the solver (source: engine coolant property database [8]):

T (°C)	EG 30 % ρ (kg/m ³)	EG 50 % ρ (kg/m ³)	EG 70 % ρ (kg/m ³)
0	1060	1090	1130
20	1048	1082	1115
40	1035	1070	1100
60	1020	1055	1085
80	1005	1040	1070
100	990	1025	1055

3.4.6 Other Fluid & Inlet Parameters

Parameter	Default	Range	Notes
β_1	20°	[5, 80]°	Inlet blade angle. Used for inlet velocity triangle and incidence only. Does not affect D_2 iteration.
u_{2_max}	200 m/s	> 0	Max tip speed. Solver aborts if u_2 exceeds this. Typical steel limit: 150–250 m/s.
D_1	0 (auto)	≥ 0	Inlet diameter. 0 = auto ($0.5 \cdot D_2$).
b_1	0 (auto)	≥ 0	Inlet width. 0 = auto via V_{m1} target.
Vu_1	0 m/s	—	Inlet pre-swirl. 0 = no pre-swirl.
V_{m1} target	3.5 m/s	[0.5, 10]	Target meridional inlet velocity for auto b_1 . Typical: 2–6 m/s.

3.5 Solver Settings (Advanced)

Parameter	Default	Range	Notes
α (relaxation)	0.3	[0.05, 1.0]	Successive relaxation factor. $D_{2_new} = D_{2_old} \cdot (1 + \alpha \cdot e)$. Reduce if solver oscillates.
Tolerance	0.01	> 0	Convergence criterion: $ e = (H_{net} - H_{target})/H_{target} < tol$.
Max iterations	200	≥ 10	Maximum Newton-relaxation iterations before declaring non-convergence.

4. Reading the Results

4.1 Geometry & Performance Metrics

Output	Symbol	Unit	Description
Outer diameter	D_2	m	Converged impeller outer diameter.
Outlet width	b_2	m	Outlet width: $b_2 = Q/(\pi \cdot D_2 \cdot V_{m2})$.
Width ratio	b_2/D_2	—	Design guideline: 0.02–0.25.
Blade tip speed	u_2	m/s	$u_2 = \pi \cdot D_2 \cdot n/60$.
Meridional velocity	V_{m2}	m/s	$V_{m2} = k_m \cdot u_2$.
Theoretical whirl	V_{u2_th}	m/s	$V_{u2_th} = u_2 - V_{m2}/\tan(\beta_2)$.
Actual whirl	V_{u2_act}	m/s	$V_{u2_act} = \sigma \cdot V_{u2_th}$.
Net head	H_{net}	m	Delivered net head (converged to H_{target}).
Specific speed (design)	ns_{design}	—	Specific speed from input Q, H, n (dimensionless, SI).
Specific speed (deliv.)	$ns_{deliv.}$	—	Specific speed from converged geometry.
Shape number	Ω_s	—	Dimensionless specific speed.
Shaft power	P_{shaft}	kW	$P_{shaft} = \rho \cdot g \cdot Q \cdot H_{net}/\eta$.
Slip factor	σ	—	Converged slip factor.
Euler head	H_{euler}	m	$H_{euler} = (u_2 \cdot V_{u2_act} - u_1 \cdot V_{u1})/g$.
Hydraulic efficiency	η_h	—	$\eta_h = H_{net}/H_{euler}$.

4.2 Outlet Velocity Triangle

The velocity triangle panel shows horizontal bars proportional to u_2 , V_{m2} , V_{u2_th} , and V_{u2_act} — an immediate visual check of flow kinematics at the impeller exit.

4.3 Inlet Velocity Triangle

Reports D_1 , b_1 , V_{m1} , u_1 , W_1 (relative inlet velocity), and incidence angle ($\beta_{1_flow} - \beta_{1_blade}$). A large incidence indicates a mismatch between the blade angle and the incoming relative flow direction, which typically increases losses.

4.4 Head & Losses

Breaks down the total head budget into Euler head, hydraulic friction loss ($k_h \cdot H_{euler}$), incidence loss ($k_i \cdot (V_{m1}^2/2g) \cdot \text{incidence}^2$), and disk-friction loss ($k_{df} \cdot u_2^2/(2g) \cdot D_2$). The sum equals H_{net} .

4.5 Efficiency

Reports the hydraulic efficiency η_h (internal) and the overall η . If η was provided as a non-zero input, the reported η equals the input and no loss model is active. If $\eta = 0$, the physics-based model is used and η_h is computed from the loss budget.

4.6 Specific Speed

Reports both ns_{design} and $ns_{\text{delivered}}$. A significant difference between the two indicates internally inconsistent input assumptions — design parameters should be revised.

The dimensionless (SI) specific speed used throughout the solver is:

$$\Omega s = \omega \cdot Q^{0.5} / (g \cdot H)^{0.75}$$

where $\omega = 2\pi n/60$ (rad/s), Q in m^3/s , $g = 9.81 \text{ m/s}^2$, H in m . The European ns (rpm, m^3/s , m) equals $\Omega s \times 52.93$; the US customary Ns (rpm, US gal/min, ft) equals $\Omega s \times 2733$.

4.7 km Sensitivity & Consistency

$dD_2/dk_m \approx -0.5 \cdot D_2/k_m$: a 10 % increase in k_m decreases D_2 by approximately 5 %. This card also checks whether the chosen k_m falls within the Gülich (2020) recommended range for the computed ns_{design} regime.

4.8 Validation Checks

Four design guideline checks are displayed:

- β_2 in $[15, 45]^\circ$
- b_2/D_2 in $[0.02, 0.25]$
- ns_{design} in $[10, 80]$
- σ in $[0.65, 0.97]$

A PASS badge is shown in green; FAIL in red. Failed checks do not prevent the solver from reporting results, but the design should be revised.

4.9 Iteration Log

Displays the Newton-relaxation history: iteration number, current D_2 , σ , H_{net} , and relative error. Useful for diagnosing slow convergence or oscillation. If the log shows divergence, reduce the relaxation factor α .

5. Performance Curves

① Frozen geometry behaviour: Each time Compute Curve is clicked, the solver re-runs the sweep using the currently converged geometry and replaces the existing chart. If you run a new design case (Run Case), the geometry changes and the previous curve is lost — the chart always reflects the most recently computed case.

After a successful design-point solve, the Performance Curves panel appears below the results. This module freezes the converged impeller geometry (D_2 , b_2 , D_1 , b_1 , n , β_2 , Z , σ model) and sweeps flow rate Q across a user-defined range to compute off-design performance.

⚠ Off-Design Modelling Assumptions: The performance curve module operates under the following simplifying assumptions: Impeller geometry is frozen (D_2 , b_2 , β_2 , Z , n are held constant across the entire Q range). The slip factor σ is re-evaluated at each flow point using the selected slip model, but the underlying geometric parameters are unchanged. Loss coefficients k_h , k_i , and k_{df} are held constant across the Q sweep — they do not adapt to off-design flow conditions. The model does not account for: flow separation at off-design incidence; secondary flows and recirculation; or volute–impeller interaction losses. Consequently, predicted curves are indicative of design-neighbourhood behaviour and should not be treated as test-accurate performance maps.

5.1 Controls

Parameter	Default	Range	Notes
Q span — lower bound	0.3	[0.05, 0.99]	Lower bound as a fraction of Q_{design} (e.g. 0.3 = 30 % of design flow).
Q span — upper bound	1.4	[1.01, 2.5]	Upper bound as a fraction of Q_{design} .
Points	30	[5, 100]	Number of flow-rate points in the sweep.

5.2 Available Curves

- **$H(Q)$** — Net head vs. flow rate. Design point marked with vertical dashed line.
- **$\eta(Q)$** — Efficiency vs. flow rate. BEP (Best Efficiency Point) marked with a filled circle.
- **$P(Q)$** — Shaft power vs. flow rate in kW.
- **$\text{Incidence}(Q)$** — Inlet incidence angle vs. flow rate. Positive = underflow; negative = overload.

① BEP is defined as the flow rate where η is maximum. For well-designed impellers, BEP should align closely with the design-point Q . A significant offset suggests design parameters should be revised.

5.3 Performance Table

Below the chart, a tabular summary lists Q , H_{net} , η , P_{shaft} , σ , k_m , Incidence, H_{euler} , H_{hyd} , and H_{shock} for every computed point. The BEP row is highlighted in green. CSV export includes only design-point results, not the curve sweep.

6. Multi-Case Workflow

The solver supports up to 8 simultaneous design cases, enabling rapid parametric studies (e.g. varying n , β_2 , or the slip model) without re-entering data.

6.1 Adding Cases

- Enter the desired number of cases (1–8) in the Step 1 field and click Apply.
- Tabs labelled Case 1, Case 2, ... appear. Click a tab to switch to that case.
- From Case 2 onwards, a 'Copy inputs from' dropdown allows duplicating any existing case as a starting point.

6.2 Running Cases

- **Run Case** — solves only the currently active (displayed) case.
- **Run All Cases** — solves all cases sequentially. Tabs turn green (converged) or red (failed).
- **Reset All** — clears all results and returns all cases to their input state.

6.3 Case Comparison Table

Once at least two cases are solved, a Case Comparison table appears below the results. Key design parameters and outputs are listed side by side for all solved cases. The best value in each row is highlighted in green; the worst in red.

6.4 CSV Export

Click Export CSV to download a comma-separated file containing the design-point results for all converged cases. The file includes geometry, performance, velocity triangle, specific speed, and validation results.

Column	Description
Case	Case number (1–8)
Q_m3s	Flow rate [m³/s]
H_target_m	Target net head [m]
n_rpm	Rotational speed [rpm]
SlipModel	Slip model used (stodola / wiesner / pfleiderer / const)
sigma	Converged slip factor [—]
D2_m	Outer diameter [m]
b2_m	Outlet width [m]
b2_D2	Width ratio [—]
u2_ms	Blade tip speed [m/s]
Vm2_ms	Meridional velocity at outlet [m/s]
Vu2_ms	Actual whirl component [m/s]
H_euler_m	Euler head [m]
H_net_m	Converged net head [m]
P_shaft_kW	Shaft power [kW]
eta_h	Internal hydraulic efficiency [—]
ns_design	Specific speed from inputs [—, SI]
ns_delivered	Specific speed from geometry [—, SI]
Omega_s	Dimensionless shape number [—]
D1_m	Inlet diameter [m]
b1_m	Inlet width [m]
Vm1_ms	Meridional inlet velocity [m/s]
incidence_deg	Inlet incidence angle [°]
b2D2_pass	b ₂ /D ₂ validation: PASS / FAIL
ns_pass	ns _{design} validation: PASS / FAIL
sigma_pass	σ validation: PASS / FAIL
beta2_pass	β ₂ validation: PASS / FAIL

7. Troubleshooting

"Solver did not converge"

The iteration reached the maximum step count without $|e| < \text{tol}$. Try the following:

- Reduce relaxation factor α (e.g. from 0.3 to 0.15 or 0.10).
- Increase Max iterations (e.g. to 400 or 600).
- Switch slip model — Stodola is the most numerically stable.
- Check that km is within the recommended range for the target ns (use the km/ns Consistency panel).
- Verify that H_{target} is achievable at the specified n — very high heads may require large D_2 , which may violate the u_{2_max} limit.

"Tip speed limit exceeded"

$u_2 = \pi \cdot D_2 \cdot n / 60$ exceeded u_{2_max} . Options:

- Reduce rotational speed n .
- Increase β_2 (reduces D_2 for the same head).
- Increase the u_{2_max} limit if material and stress calculations allow it.
- Accept a lower head target H .

Wiesner fallback warning

Shown when the Wiesner validity condition $\phi < \phi_{\text{lim}}$ is violated for the chosen km and Z . The solver automatically falls back to Stodola. To suppress the warning, either switch to Stodola explicitly or reduce km .

σ out of range warning

Shown when the computed σ falls outside $[0.65, 0.97]$. Very low σ (< 0.65) indicates heavy slip and may point to an excessive blade angle or too few blades. Consider increasing Z or reducing β_2 .

Validation checks failing

Failed checks are warnings, not errors. The design results are still reported. However, values outside the stated ranges are outside the empirical basis of the correlations and should be treated with caution.

8. Physics & Assumptions

- The solver assumes zero inlet pre-swirl ($Vu_1 = 0$, $\alpha_1 = 90^\circ$) by default — i.e. no inlet guide vanes.
- 1D meanline: flow properties are uniform across each meridional cross-section.
- The impeller is treated as a single-stage, single-suction centrifugal stage.
- Hydraulic losses are modelled with three lumped coefficients (k_h , k_i , k_{df}). The defaults are representative of well-designed industrial impellers but will differ from any specific machine.
- Mechanical losses (bearing friction, seal drag, balance disc leakage) are not included.
- Cavitation / NPSH assessment is not performed.
- Volute, diffuser, and return-channel losses are not modelled.

8.1 Loss Model Formulas ($\eta = 0$ mode)

When $\eta = 0$, the solver computes hydraulic efficiency from first principles using three loss terms, each parameterised by one coefficient:

Loss Term	Formula	Notes
Hydraulic friction	$H_f = k_h \cdot H_{\text{euler}}$	Proportional to Euler head. Represents passage friction, leakage, and recirculation in the impeller channels.
Incidence loss	$H_i = k_i \cdot (Vm_1^2/2g) \cdot i^2$	i = incidence angle in radians ($\beta_{1_blade} - \beta_{1_flow}$). Zero at design incidence; peaks away from design flow.
Disk friction	$H_{df} = k_{df} \cdot u_2^2/(2g) \cdot D_2$	Acts on the power side: $P_{df} = \rho \cdot g \cdot Q \cdot H_{df}$. Sensitive to tip speed; largely independent of flow rate.

$$H_{\text{net}} = H_{\text{euler}} - H_f - H_i$$

$$P_{\text{shaft}} = P_{\text{useful}} + P_{df} = \rho \cdot g \cdot Q \cdot H_{\text{net}} + \rho \cdot g \cdot Q \cdot H_{df}$$

① For complete pump system analysis, results from this tool should be combined with volute/diffuser design, NPSH calculations, mechanical design, and rotordynamic analysis.

8.2 Model Validity Envelope

The 1D meanline model is calibrated for typical industrial centrifugal pump duty. Accuracy degrades outside the following bounds and the user should interpret results with increased conservatism.

Parameter	Reliable range	Limitation outside range
Specific speed ns (SI)	10 – 80	Below $ns \approx 10$: viscous and recirculation effects dominate; loss model underpredicts. Above $ns \approx 80$: mixed- or axial-flow geometry assumed; radial-impeller correlations inapplicable.
Reynolds number $Re = \frac{u_2 \cdot D_2}{\nu}$	$Re > 10^5$ (turbulent)	At low Re (high viscosity, small impeller, or low speed), friction factors increase significantly and the hydraulic efficiency η_h predicted by kh will be optimistic.
Dynamic viscosity μ	$\mu \approx 0.001$ Pa·s (water-like)	High-viscosity fluids ($\mu > \sim 5$ mPa·s) require viscosity corrections per Gülich [4] §13. The current loss model — particularly kdf — does not scale with μ .
Flow rate ratio Q/Q_{design}	0.5 – 1.3 (near BEP)	Strong off-design operation introduces separation, recirculation, and rotating stall. Predicted curves are indicative only beyond ± 30 % of design flow.
Impeller type	Radial (shrouded) single-stage	Mixed-flow, open, or multistage configurations are outside the model scope.

ⓘ The accuracy of this tool is comparable to classical preliminary design methods (cf. Stepanoff, Pfleiderer). It is not a substitute for test data or CFD-validated performance maps.

8.3 When to Use Higher-Fidelity Analysis (CFD)

The following conditions indicate that 3D CFD or experimental testing is required to achieve reliable results:

- High blade loading (low Z , large β_2 , or high ns approaching mixed-flow territory).
- Strong off-design operation ($Q/Q_{\text{design}} < 0.5$ or > 1.5), where separation and recirculation are probable.
- Cavitation / NPSH-critical applications where suction geometry and vapour bubble dynamics must be resolved.
- Complex inlet conditions: pre-swirl from upstream components, non-uniform velocity distribution, or pipe-induced distortion.
- Detailed loss breakdown required for impeller-volute interaction, secondary flow losses, or leakage path quantification.
- High-viscosity fluids ($\mu > \sim 5$ mPa·s) where Reynolds-number corrections are non-trivial.

ⓘ FlowForge v1.0 provides the design-point geometry and velocity triangles that serve as the starting point for CFD meshing and boundary condition setup. It is not a replacement for higher-fidelity methods.

9. References

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